

- 2 Natural selection and selective breeding (artificial selection) are processes that result in changes in the gene pool of a population.

Natural selection and selective breeding have implications for humans.

- (a) If a person with a bacterial infection does not finish the course of an antibiotic given, it provides the conditions for a population of bacteria to become resistant to this antibiotic.

- (i) A mutation in a bacterial gene can give resistance to an antibiotic. Directional selection can occur when the antibiotic is present in the environment.

A bacterium can also gain resistance when it receives genetic material from another bacterium in a process known as horizontal gene transfer.

Outline how directional selection and horizontal gene transfer result in a new population of bacteria that is resistant to an antibiotic.

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- (ii) Some bacterial diseases can be treated only with one antibiotic, because the bacterial pathogens are resistant to all other antibiotics.

A drug is being developed to help treatment.

- The drug is a small polynucleotide.
- The drug inhibits translation of the messenger RNA (mRNA) produced by transcription of the gene associated with antibiotic resistance.
- The bacteria are then susceptible to more antibiotics.

Suggest **and** explain how the drug could cause bacteria to become susceptible to more antibiotics.

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(b) (i) Outline how natural selection differs from selective breeding.

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(ii) Selective breeding is used to produce uniform varieties of maize. The maize plants in a crop ripen at the same time and are the same height. The advantage of this is that harvesting is easy and quick. The disadvantage is that farmers must buy new seeds each year.

Explain why farmers must buy new seeds each year.

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